

# The ODE Template and Its Domain Equivalents

## Mechanical and electrical forms of one second-order equation

Hob Nilre & Bo C. Herlin

2026-10-08

### Abstract

Three linear relations and a balance give  $a\ddot{y} + b\dot{y} + cy = f$ . We derive its mechanical and electrical forms, the unit conversions between them, and the reciprocal map that exchanges common and summed quantities. Harmonic substitution gives the corresponding terminal ratio.

PDF created: 2026-10-08 13:07:40 UTC

Latest on GitHub: <https://github.com/hobnilre/physics-ode-template>

## 1 The common template

Let  $a, b, c > 0$  be constants and  $D = d/dt$ ; dots and primes also denote time derivatives. For a common quantity  $u$ , define  $\dot{y} = u$ ,  $h = au$  and three terms of the same dimension:

$$z_a = \dot{h} = a\dot{u}, \quad z_b = bu, \quad z_c = cy. \quad (1)$$

Their signed balance  $z_a + z_b + z_c = 0$  gives

$$\boxed{a\ddot{y} + b\dot{y} + cy = 0.} \quad (2)$$

The terms depend on the rate of  $u$ , on  $u$  itself, and on its accumulated coordinate. Their physical meanings determine the domain.

With time dimension  $T$ , the defining relations require

$$[y] = [u]T, \quad [h] = [z]T, \quad [a] = \frac{[z]T}{[u]}, \quad [b] = \frac{[z]}{[u]}, \quad [c] = \frac{[z]}{[u]T}. \quad (3)$$

Here  $[z]$  is the common term dimension. The equivalent first-order form is

$$Dy = \frac{h}{a}, \quad Dh = -cy - \frac{b}{a}h, \quad (4)$$

with initial coordinates  $y(t_0)$  and  $h(t_0) = a\dot{y}(t_0)$ .

### 1.1 Forcing and harmonic form

A prescribed sum gives  $P_2(D)y = f$ , where  $P_2(s) = as^2 + bs + c$ . For  $x(t) = \text{Re}(\hat{x}e^{i\omega t})$ ,  $i^2 = -1$  and  $\omega > 0$ , differentiation multiplies the phasor by  $i\omega$ . Hence

$$\hat{f} = P_2(i\omega)\hat{y} = \left(ai\omega + b + \frac{c}{i\omega}\right)\hat{u}, \quad \hat{u} = i\omega\hat{y}. \quad (5)$$

This describes the harmonic component; the full solution also needs its initial coordinates. *ODE Coefficient Synthesis*, Sections 1–3, extends the dimension rule to coefficients of higher powers of  $D$ .

## 2 Four domain forms

In mechanics use mass  $m$ , damping  $d$ , stiffness  $k$ , displacement  $x$ , velocity  $v$ , force  $F$  and momentum  $p$ . In electrics use inductance  $L$ , resistance  $R$ , capacitance  $C$ , current  $i$ , voltage  $e$ , charge  $q$  and flux linkage  $\lambda$ . All parameters are positive constants.

**Table 1.** Coefficients and coordinates;  $x_s$  is spring displacement. The last row gives the quantity being summed.

| Template            | MCK   | KCM      | LRC       | CRL       |
|---------------------|-------|----------|-----------|-----------|
| $a$                 | $m$   | $1/k$    | $L$       | $C$       |
| $b$                 | $d$   | $1/d$    | $R$       | $1/R$     |
| $c$                 | $k$   | $1/m$    | $1/C$     | $1/L$     |
| Common $u$          | $v$   | $F$      | $i$       | $e$       |
| Coordinate $h = au$ | $p$   | $x_s$    | $\lambda$ | $q$       |
| Coordinate $y$      | $x$   | $p$      | $q$       | $\lambda$ |
| Type of $z_j$       | Force | Velocity | Voltage   | Current   |

### 2.1 MCK and LRC

For MCK, velocity is common and forces add:  $v = \dot{x}$ ,  $p = mv$ , and the three forces are  $\dot{p}$ ,  $dv$  and  $kx$ . For LRC, current is common and voltages add:  $i = \dot{q}$ ,  $\lambda = Li$ , and the voltages are  $\dot{\lambda}$ ,  $Ri$  and  $q/C$ . Thus

$$m\ddot{x} + d\dot{x} + kx = 0, \quad L\ddot{q} + R\dot{q} + \frac{q}{C} = 0. \quad (6)$$

These are mechanical parallel and electrical series forms. The forced LRC equation is the starting point for the signed-power calculation in *Energy Ledgers for Forced Harmonic ODEs*, Section 1.

### 2.2 KCM and CRL

For KCM, force is common and velocities add. From  $\dot{p} = F$  and  $x_s = F/k$ , the velocities are  $\dot{F}/k$ ,  $F/d$  and  $p/m$ . For CRL, voltage is common and currents add. From  $\dot{\lambda} = e$  and  $q = Ce$ , the currents are  $C\dot{e}$ ,  $e/R$  and  $\lambda/L$ . Hence

$$\frac{\ddot{p}}{k} + \frac{\dot{p}}{d} + \frac{p}{m} = 0, \quad C\ddot{\lambda} + \frac{\dot{\lambda}}{R} + \frac{\lambda}{L} = 0. \quad (7)$$

These are mechanical series and electrical parallel forms. *Interconnecting Series and Parallel ODEs*, Sections 1–2, uses the common and summed quantities to join whole blocks of either type.

## 3 Correspondence and duality

### 3.1 Changing domains

Choose positive conversion scales  $\alpha$ ,  $\beta$  and keep the time variable. The force–voltage correspondence is

$$i \leftrightarrow \alpha v, \quad e \leftrightarrow \beta F, \quad L = \frac{\beta}{\alpha} m, \quad R = \frac{\beta}{\alpha} d, \quad C = \frac{\alpha}{\beta k}. \quad (8)$$

The scales convert units and values. With  $(q, \lambda) = (\alpha x, \beta p)$  for MCK–LRC, and  $(\lambda, q) = (\beta p, \alpha x_s)$  for KCM–CRL,

$$L\ddot{q} + R\dot{q} + \frac{q}{C} = \beta(m\ddot{x} + d\dot{x} + kx), \quad (9)$$

$$C\ddot{\lambda} + \frac{\dot{\lambda}}{R} + \frac{\lambda}{L} = \alpha\left(\frac{\ddot{p}}{k} + \frac{\dot{p}}{d} + \frac{p}{m}\right). \quad (10)$$

The equations and initial coordinates therefore transform together.

### 3.2 Exchanging common and summed quantities

To construct the dual, make  $w$  common in place of the  $z_j$  and sum the corresponding  $u_j$ . Reverse the relations in (1) and their element order:

$$u_c = \frac{\dot{w}}{c}, \quad u_b = \frac{w}{b}, \quad u_a = \frac{Y}{a}, \quad \dot{Y} = w.$$

Here  $Y = au_a$ . The balance  $u_c + u_b + u_a = 0$  gives

$$\boxed{\frac{\ddot{Y}}{c} + \frac{\dot{Y}}{b} + \frac{Y}{a} = 0, \quad \mathcal{D}(a, b, c) = \left(\frac{1}{c}, \frac{1}{b}, \frac{1}{a}\right)}. \quad (11)$$

This reciprocal map sends MCK to KCM and LRC to CRL; applying it twice returns  $(a, b, c)$ .

Dividing each equation by its leading coefficient gives

$$\ddot{y} + \frac{b}{a}\dot{y} + \frac{c}{a}y = 0, \quad \ddot{Y} + \frac{c}{b}\dot{Y} + \frac{c}{a}Y = 0. \quad (12)$$

The ratio  $c/a$  is unchanged, while  $b/a$  becomes  $c/b$ . The normalized equations coincide exactly when  $b^2 = ac$ . Changing domains rescales the quantities; taking the dual exchanges what is common and what is summed.

### References

1. H. Nilre and B. C. Herlin (2026). [ODE Coefficient Synthesis](#).
2. H. Nilre and B. C. Herlin (2026). [Interconnecting Series and Parallel ODEs](#).
3. H. Nilre and B. C. Herlin (2026). [Energy Ledgers for Forced Harmonic ODEs](#).